

Forces and Free-body diagram

SS ILP Physics (Intermediate), Week 2 Class 2



Outline

- Types of forces
- Free-body diagram
- Superposition Principle

Types of forces

(a) **Weight:** gravitational force acting on a massive object due to its interaction with Earth or other massive objects. Weight is directed towards the center of the Earth or celestial body. When near the surface of the earth, $W=mg$

(b) **Contact force**

- I. Normal force: the support acting on one object in contact with another one. Normal force is perpendicular to the surface of contact.
- II. Frictional Force: the force opposing motion or the tendency of motion when two objects are in contact. Friction is directed along the contact surface.

<https://www.physicsclassroom.com/class/newtlaws/Lesson-2/Types-of-Forces>

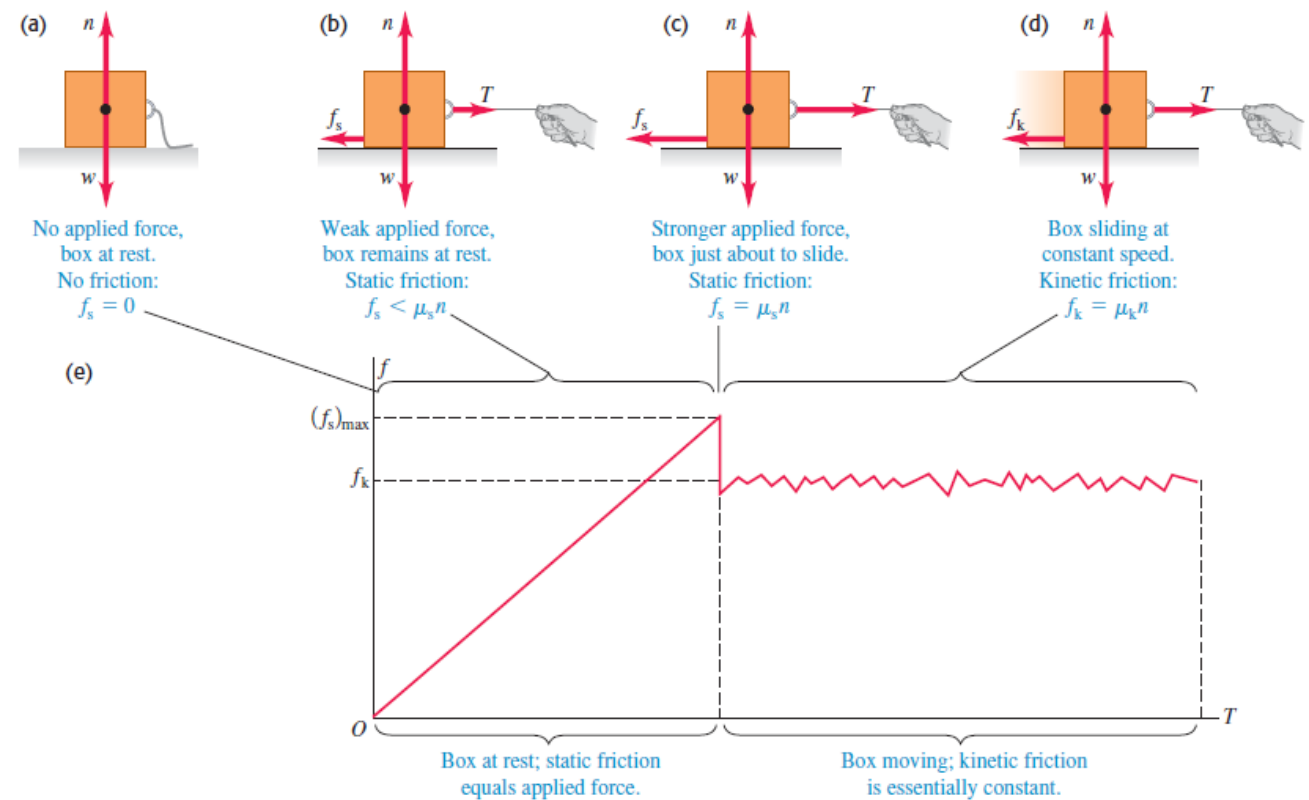
- **Static friction** ranges between 0 and $f_{s,max}$ to keep two objects in contact stationary with respect to each other. i.e.

$$0 \leq f_s \leq \mu_s N = f_{s,max}$$

μ_s is the static friction coefficient

- **Kinetic friction** is proportional to the normal contact force. i.e.

$$f_k = \mu_k N \quad \mu_k \text{ is the kinetic friction coefficient.}$$

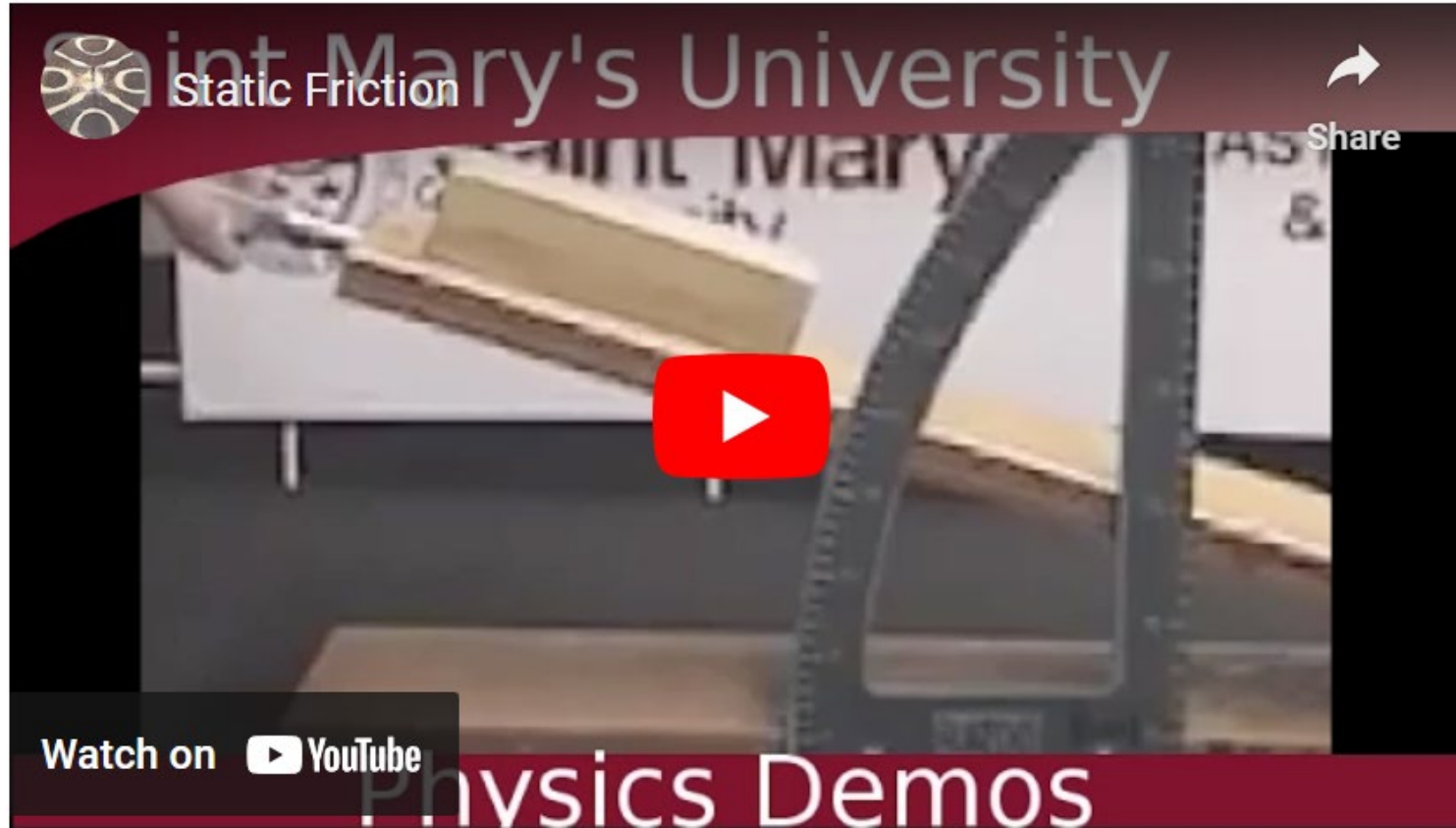


H.D. Young and R. A. Freedman, Sears and Zemansky's University Physics. 13E, Addison-Wesley, Boston, 2012.



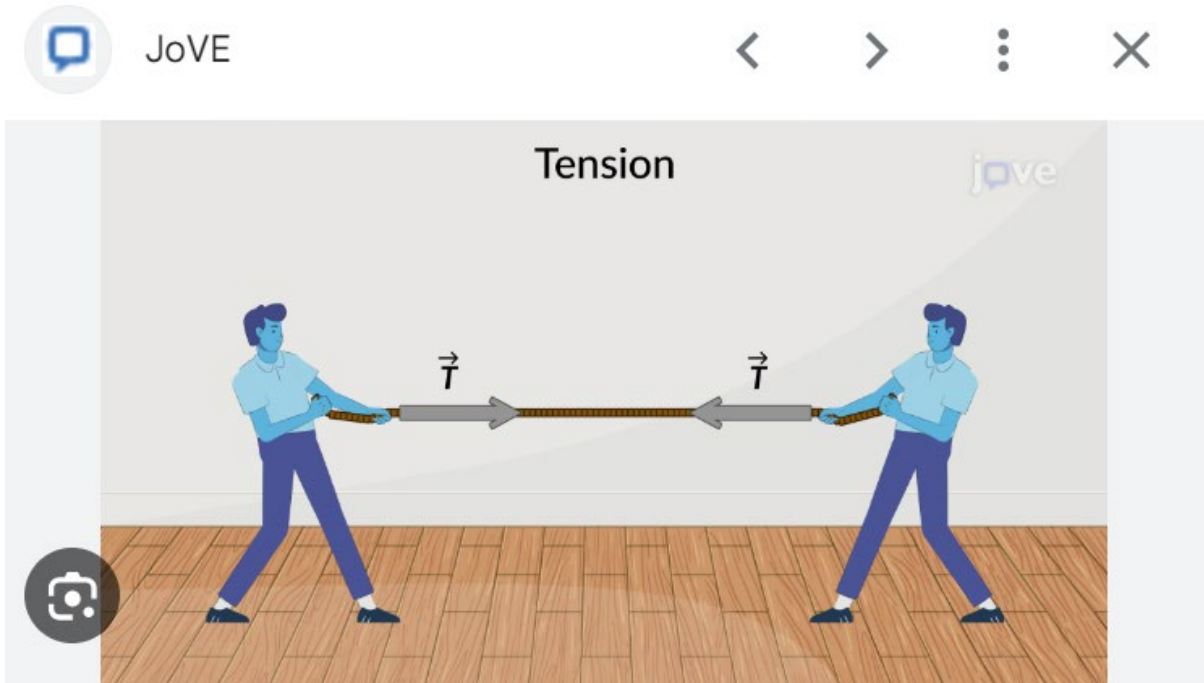
Is the friction acting on the laptop on your table now kinetic or static? How much is it? In which direction is it pointing?

Static Friction Demonstration



<https://demos.smu.ca/demos/mechanics/130-static-friction>

- c) **Tension**: the pulling force transmitted through a string or a cable when it is pulled tight. Tension is directed along the string. By assuming that ropes or strings are massless, the tension has the same magnitude throughout their length.



Pulleys are usually used to change the direction of a tension.



▲ FIGURE 6-7 A pulley changes the direction of a tension

<https://www.physicsclassroom.com/class/newtlaws/Lesson-2/Types-of-Forces>
<https://app.jove.com/science-education/v/14698/tension>

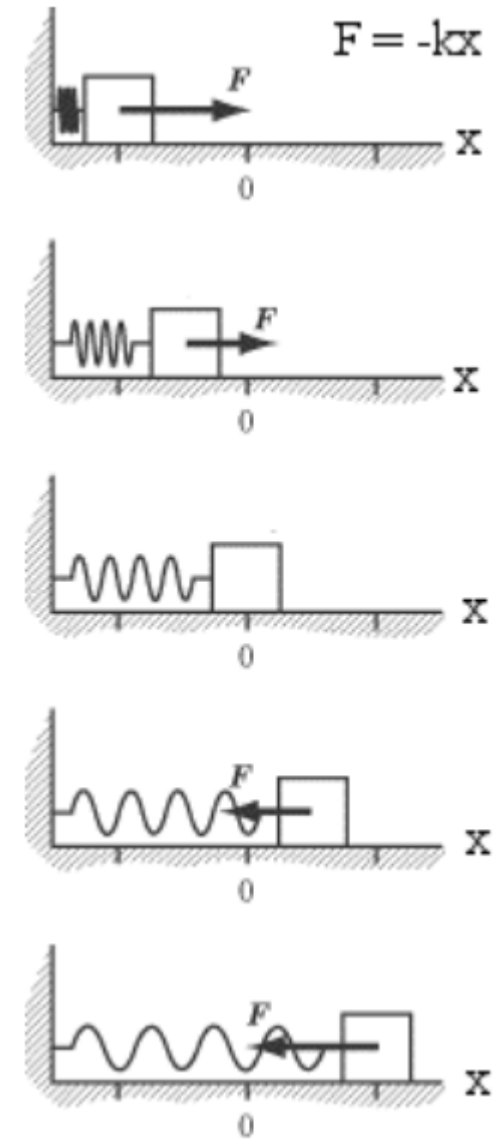
J. S. Walker, Physics, p150

(d) **Spring Force:** the force exerted by a spring when it is stretched or compressed. Spring force is directed along the spring, trying to restore it to its rest position.

Hooke's law $F = -kx$

where k is called spring constant, dependent on the property of the spring.

<https://www.physicsclassroom.com/class/newtlaws/Lesson-2/Types-of-Forces>
<https://labs.phys.utk.edu/mbreinig/phys221core/modules/m3/Hooke's%20law.html#:~:text=When%20a%20spring%20is%20stretched,spring%20to%20its%20equilibrium%20length>



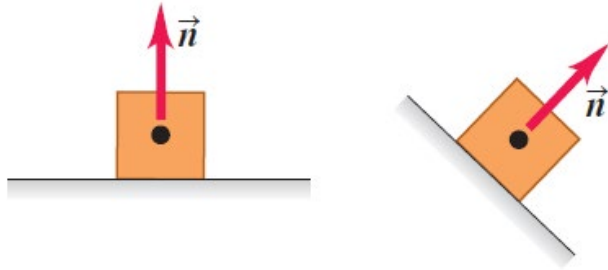
- (e) **Air Resistance:** the resistance force acting on an objects moving through the air. Air resistance opposes the motion of the object. It is usually neglected in problem analysis.
- (f) **Electrical Force:** the force acting on charged objects, through electric field.
- (g) **Magnetic Force:** the force acting on moving charged objects, through magnetic field.

<https://www.physicsclassroom.com/class/newtlaws/Lesson-2/Types-of-Forces>

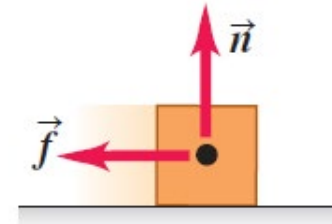
Drawing force vectors

Use a vector arrow to indicate the magnitude and direction of the force.

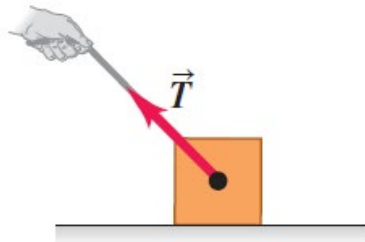
(a) **Normal force $\vec{n}$:** When an object rests or pushes on a surface, the surface exerts a push on it that is directed perpendicular to the surface.



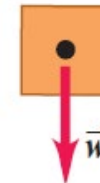
(b) **Friction force $\vec{f}$:** In addition to the normal force, a surface may exert a frictional force on an object, directed parallel to the surface.



(c) **Tension force $\vec{T}$:** A pulling force exerted on an object by a rope, cord, etc.



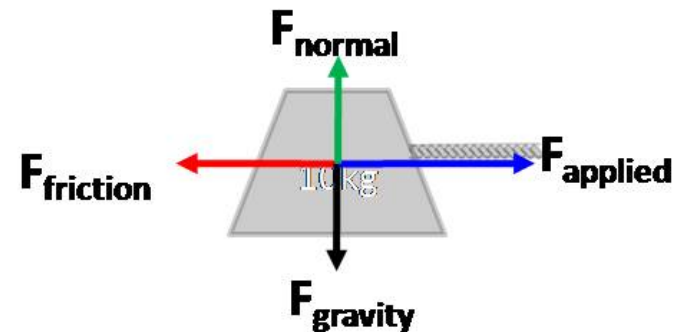
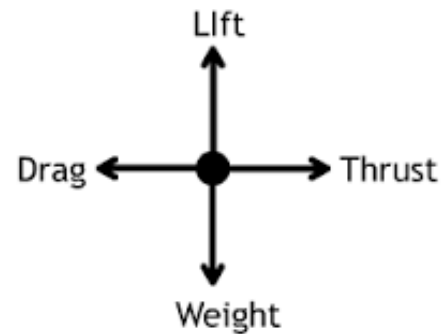
(d) **Weight $\vec{w}$:** The pull of gravity on an object is a long-range force (a force that acts over a distance).



H.D. Young and R. A. Freedman, Sears and Zemansky's University Physics. 13E, Addison-Wesley, Boston, 2012.

Free-body diagram

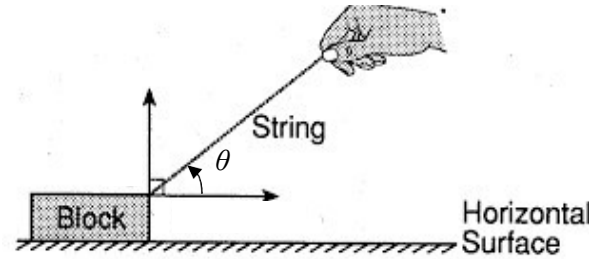
- Free-body diagrams are used to show the relative magnitude and direction of all forces acting upon an object in a given situation.
- Free body-diagrams are very useful and should be drawn first before solving the problem
- Examples:



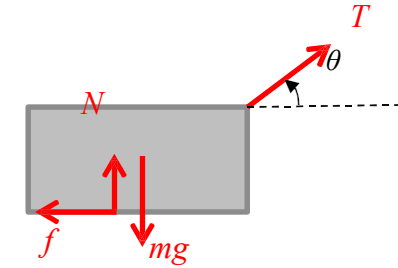
<https://math.etsu.edu/multicalc/prealpha/Chap1/Chap1-1/part4.htm>

Free-body diagram

Example: A block of mass m is pulled by a string at an angle θ from the horizontal. There is friction between the block and the surface. Identify the physical forces on the block and draw the free body diagram of the block.



https://www.ntschoools.org/cms/lib/NY19000908/Centricity/Domain/816/ap_energy_review_sc.pdf



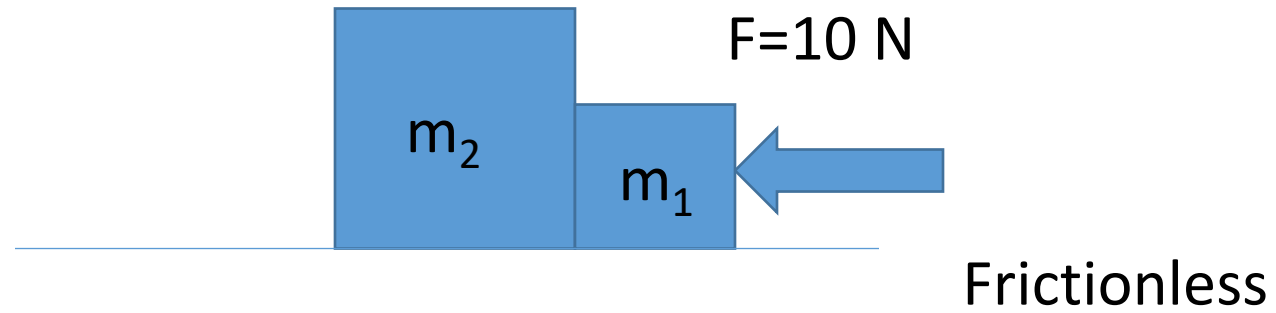
Free body diagram

Answer: four forces can be recognized.

The block

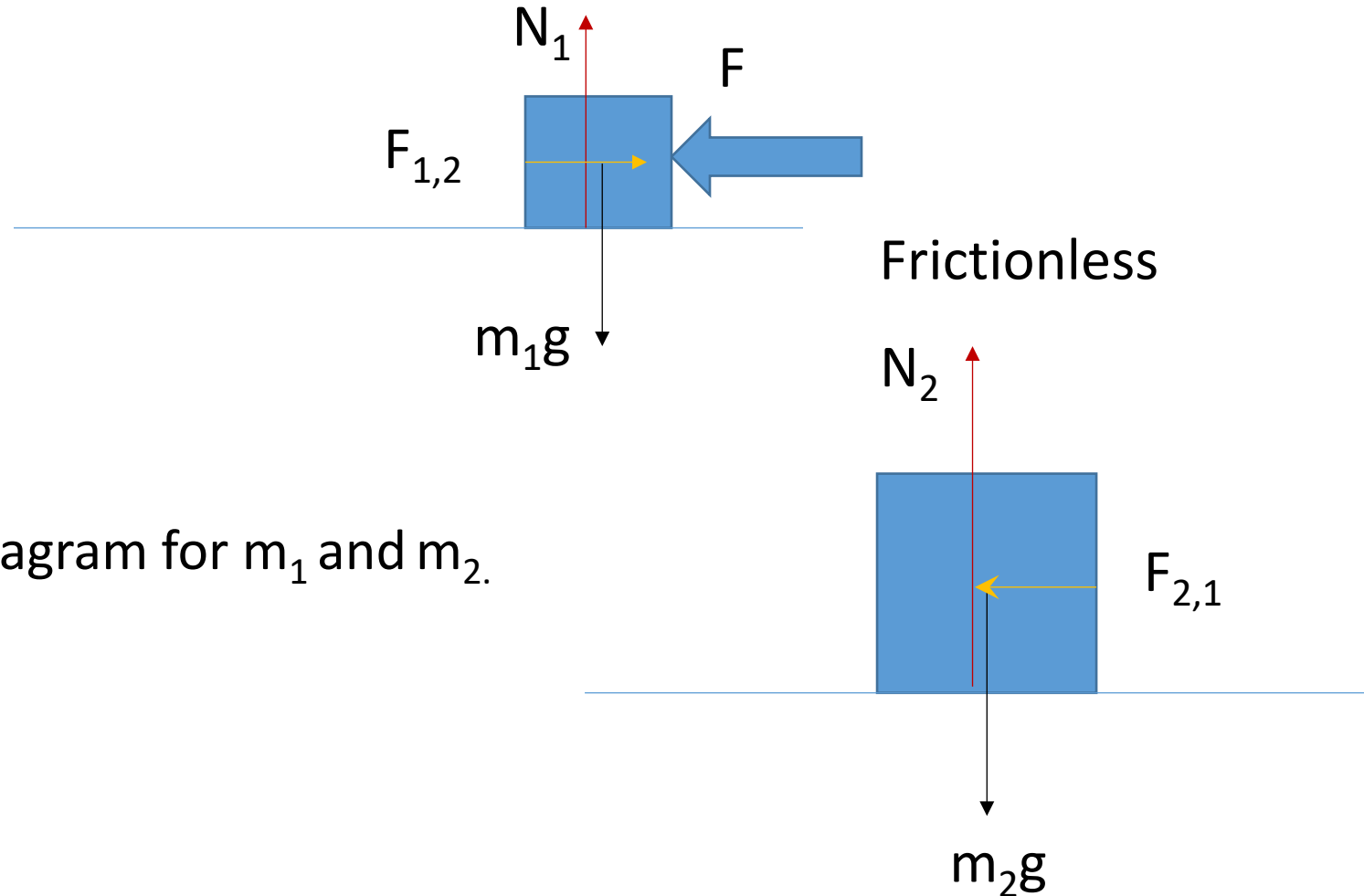
- is pulled by tension T directed at angle θ (interaction with string).
- has weight mg (gravitational attraction by the earth).
- is supported by normal force N (interaction with the surface).
- is opposed by horizontal friction force f (interaction with the surface due to relative motion or impending relative motion).

Example



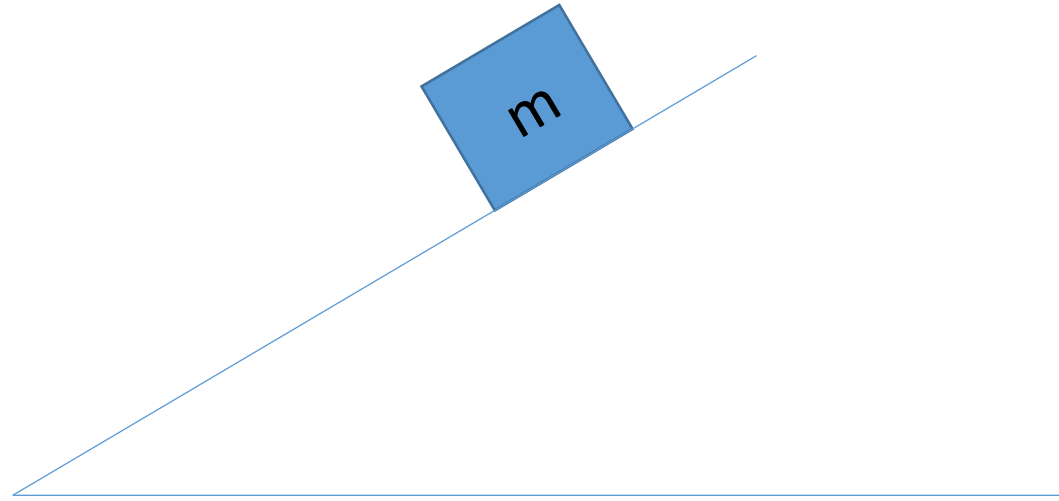
Draw free-body diagram for m_1 and m_2 .

Example: Solution



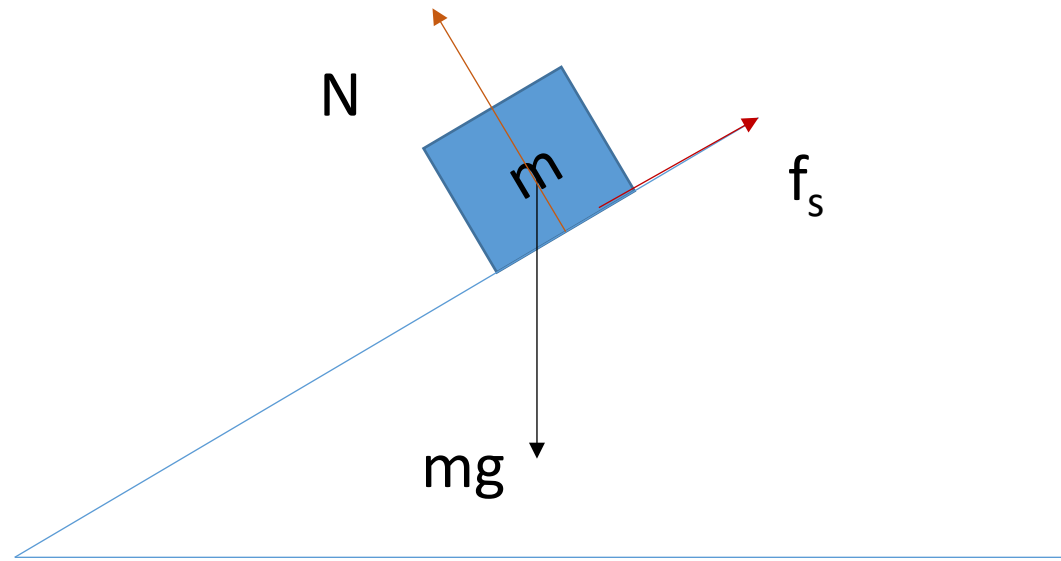
Draw free-body diagram for m_1 and m_2 .

Example



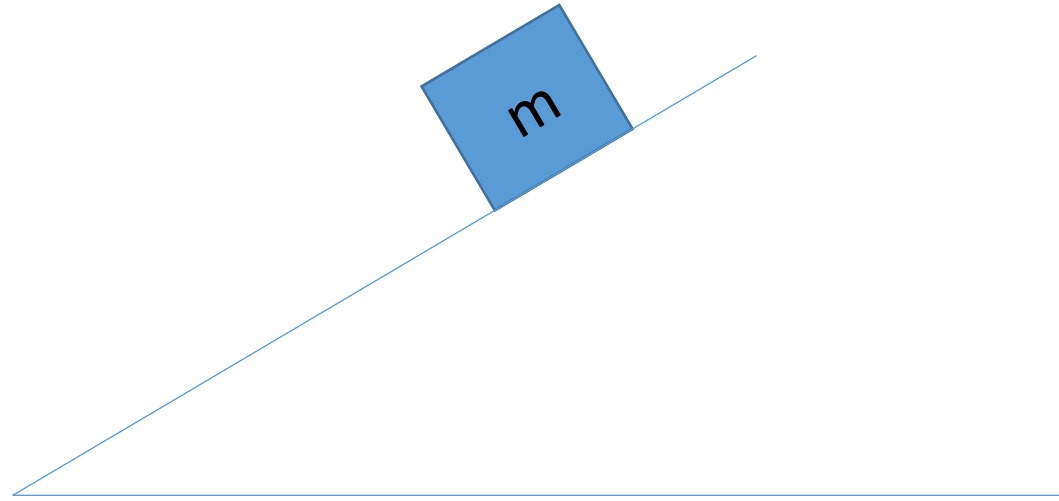
Draw free-body diagram for m which is placed on a slope, and it is at rest. Assume the surface is not frictionless.

Example: Solution



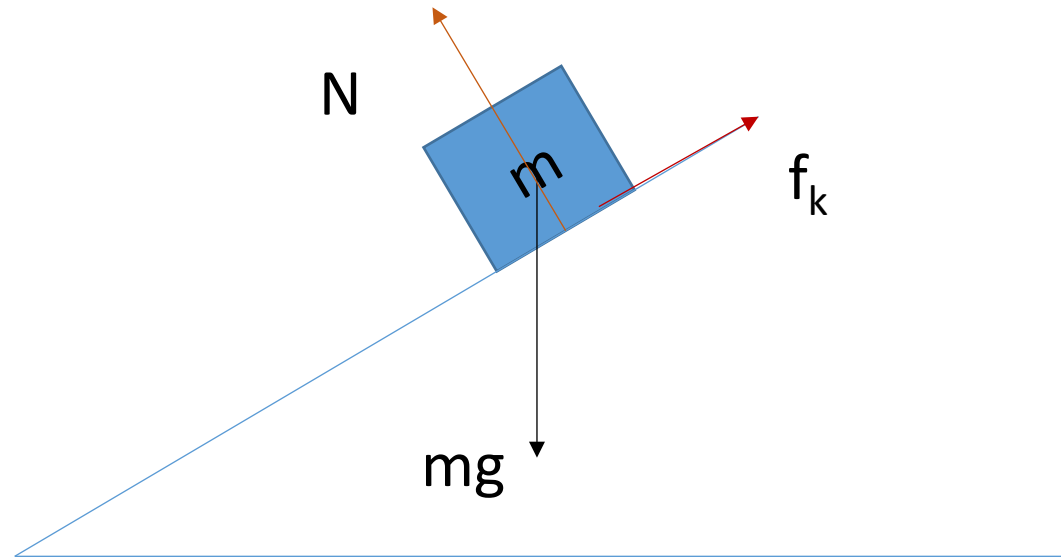
Draw free-body diagram for m which is placed on a slope, and it is at rest. Assume the surface is not frictionless.

Case Problem



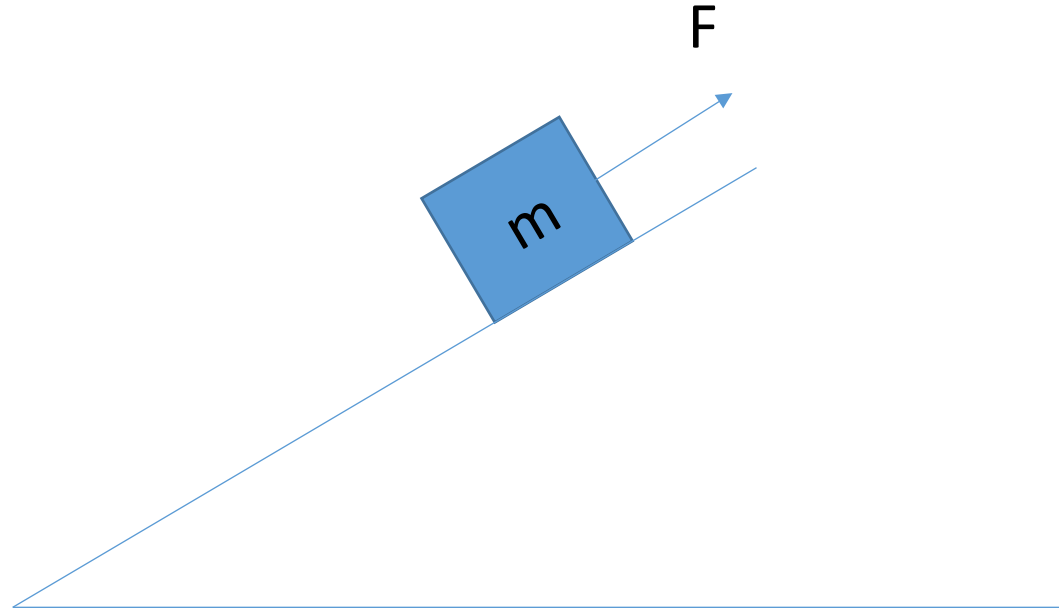
Draw free-body diagram for m which is placed on a slope, and it is moving downwards. Assume the surface is not frictionless.

Case Problem: Solution



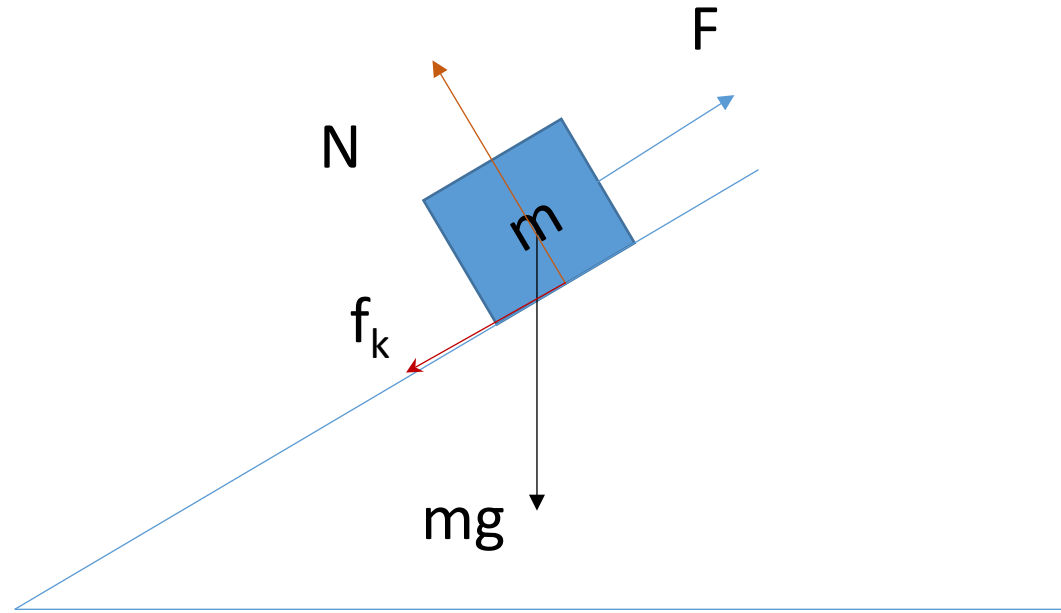
Draw free-body diagram for m which is placed on a slope, and it is moving downwards. Assume the surface is not frictionless.

Case Problem



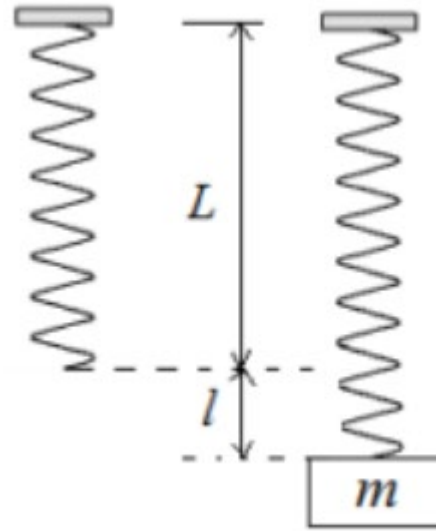
Draw free-body diagram for m which is placed on a slope, and it is moving upwards. Assume the surface is not frictionless.

Case Problem: Solution



Draw free-body diagram for m which is placed on a slope, and it is moving upwards. Assume the surface is not frictionless.

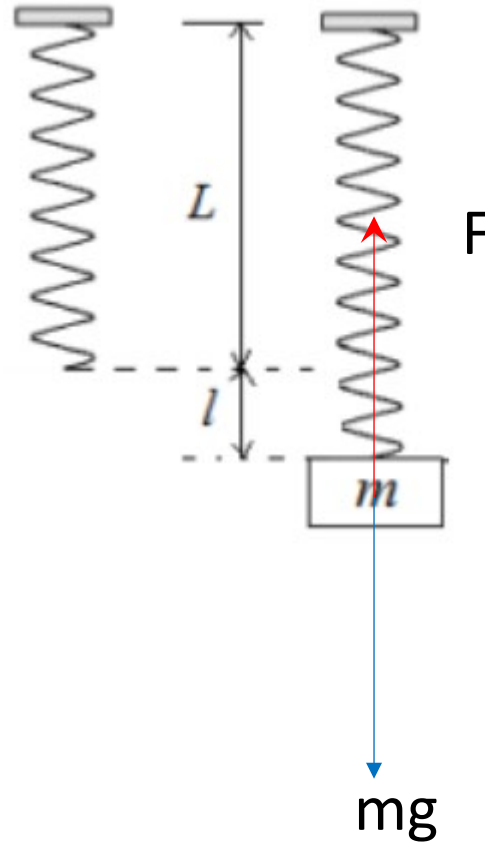
Case Problem



Draw free-body diagram for m which is suspended by a spring with length L , and it is at rest.

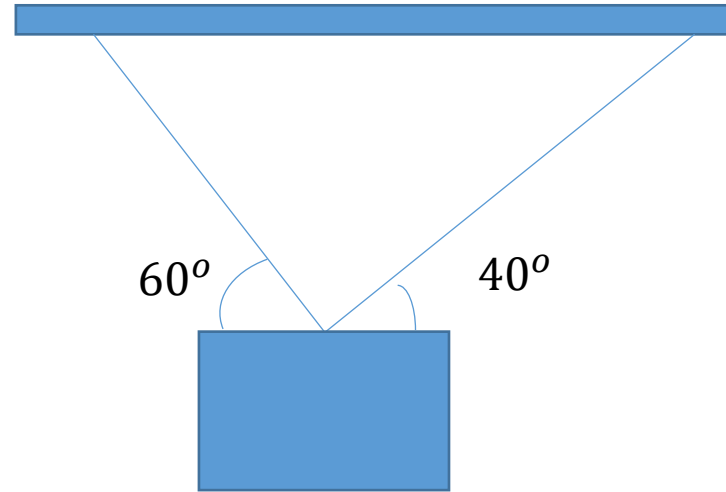
https://thefactfactor.com/facts/pure_science/physics/the-vibration-of-vertical-spring/9249/

Case Problem: Solution



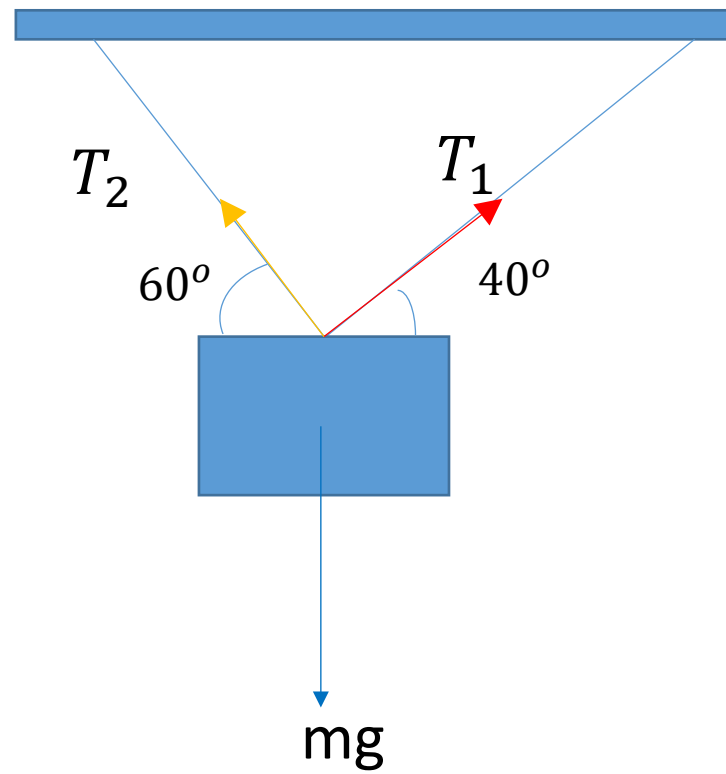
https://thefactfactor.com/facts/pure_science/physics/the-vibration-of-vertical-spring/9249/

Case Problem

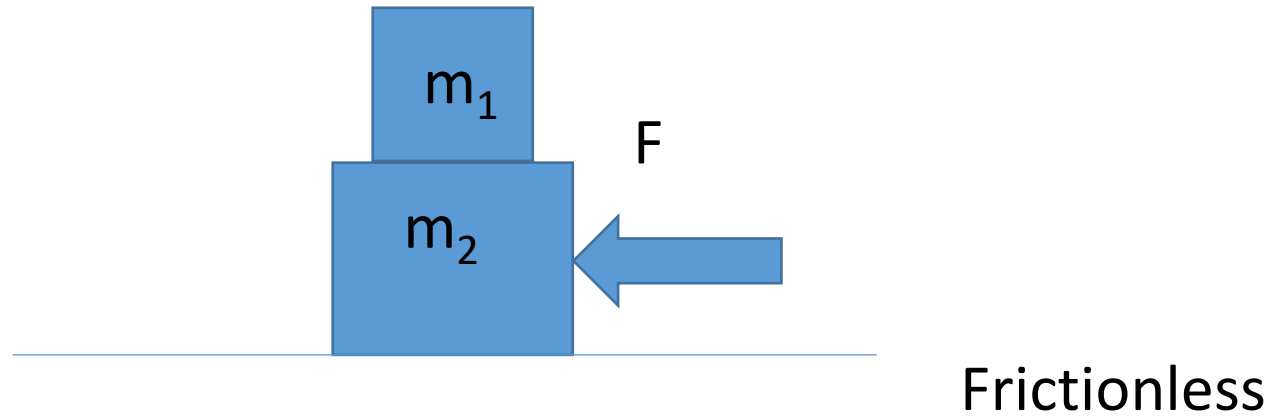


Draw free-body diagram for a block of mass m which is suspended through two cables to the top surface, and it is at rest.

Case Problem: Solution

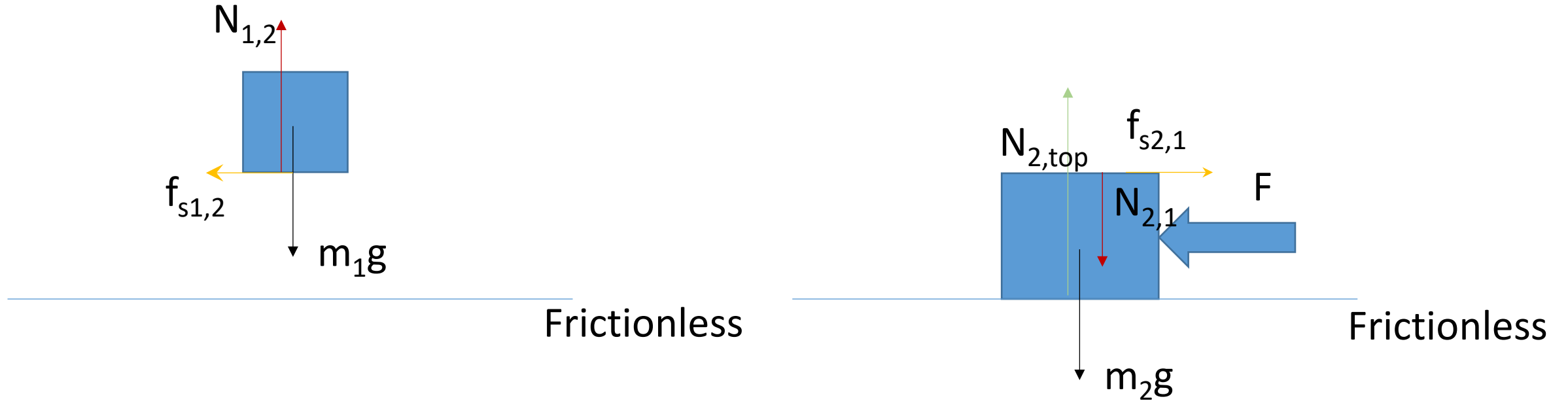


Example



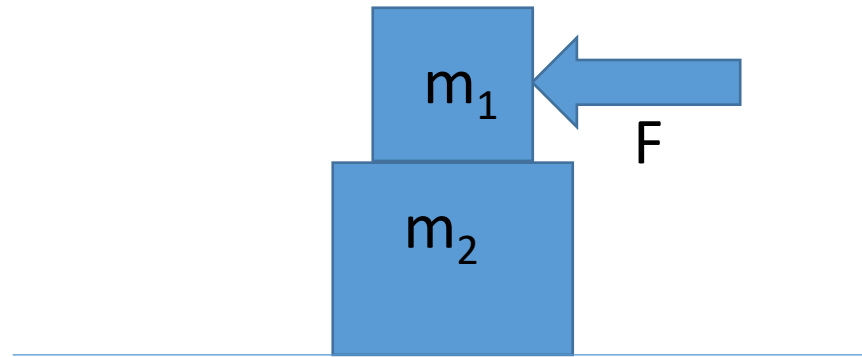
Draw free-body diagram for m_1 and m_2 that are moving to left (no relative motion between them). Assume the contact surface between m_1 and m_2 is not frictionless.

Example: Solution



Draw free-body diagram for m_1 and m_2 that are moving to left (no relative motion between them). Assume the contact surface between m_1 and m_2 is not frictionless.

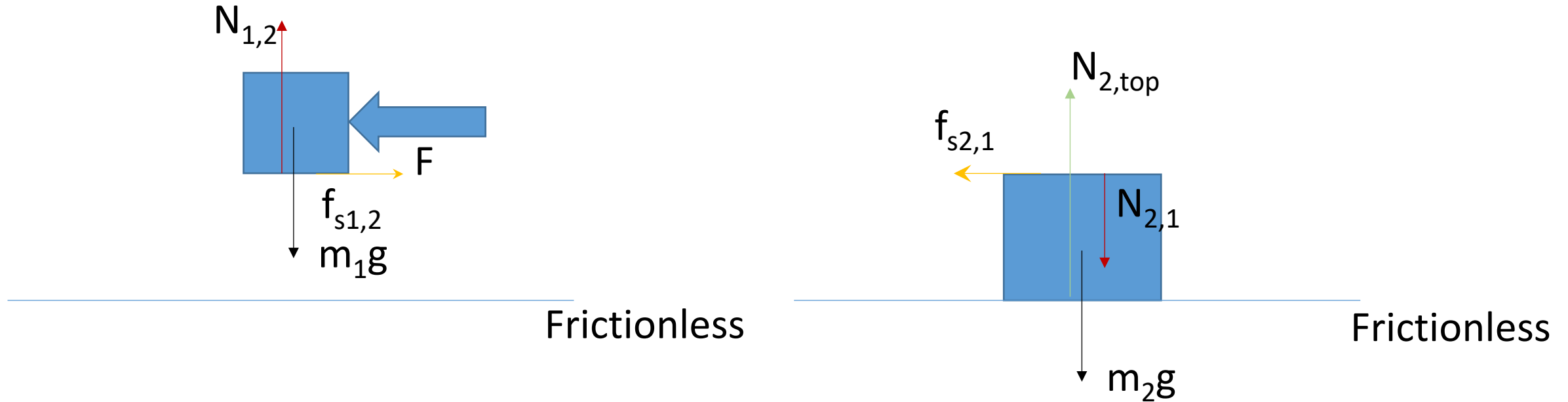
Case Problem



Frictionless

Draw free-body diagram for m_1 and m_2 that are moving to left (no relative motion between them). Assume the contact surface between m_1 and m_2 is not frictionless.

Case Problem: Solution

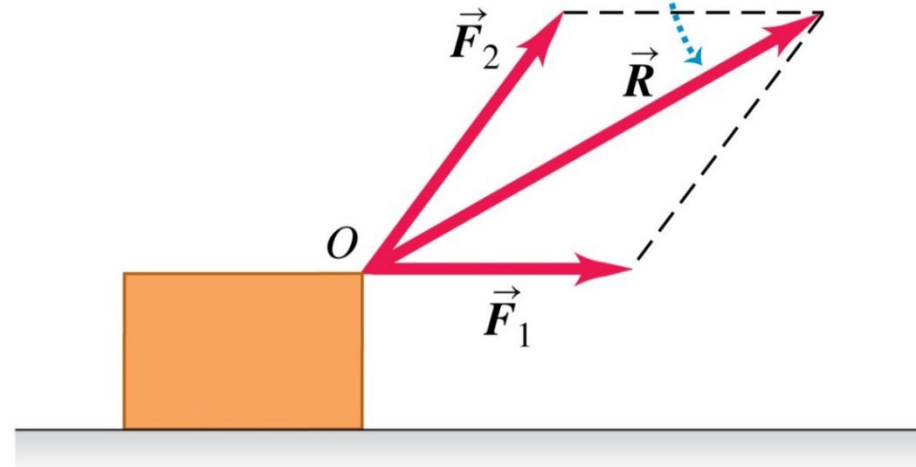


Draw free-body diagram for m_1 and m_2 that are moving to left (no relative motion between them). Assume the contact surface between m_1 and m_2 is not frictionless.

Superposition of forces

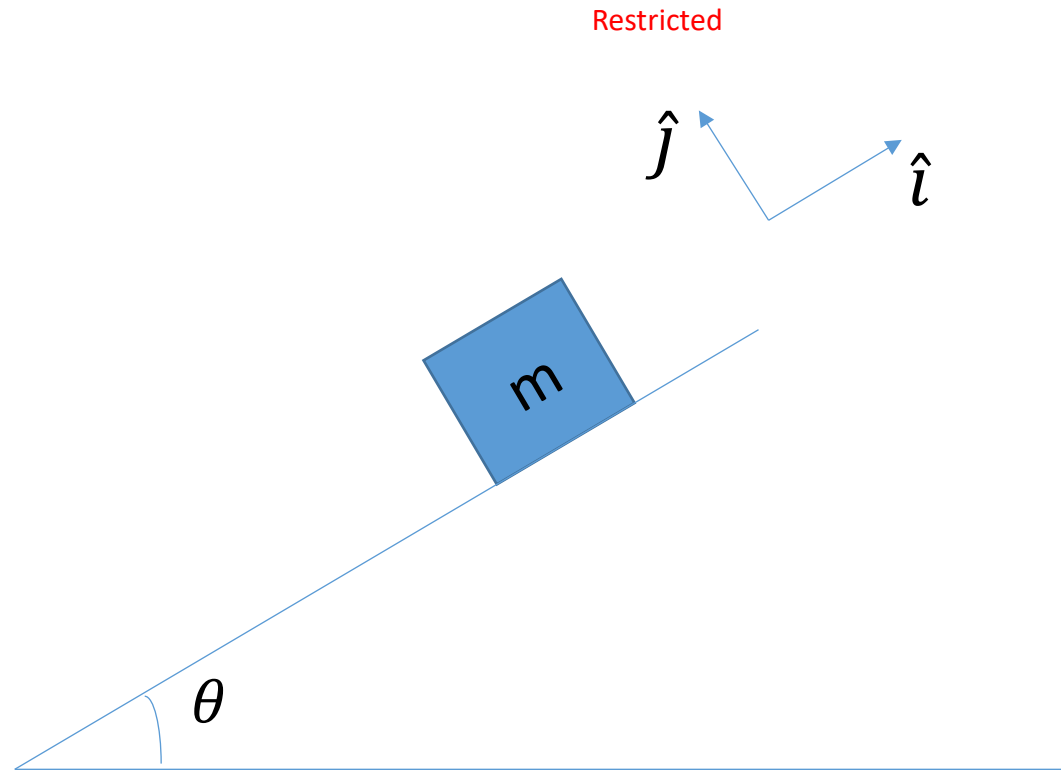
Several forces acting at a point on an object have the same effect as their vector sum (net Force or resultant force) acting at the same point.

Two forces $\vec{F}_1$ and $\vec{F}_2$ acting on a body at point O have the same effect as a single force $\vec{R}$ equal to their vector sum.



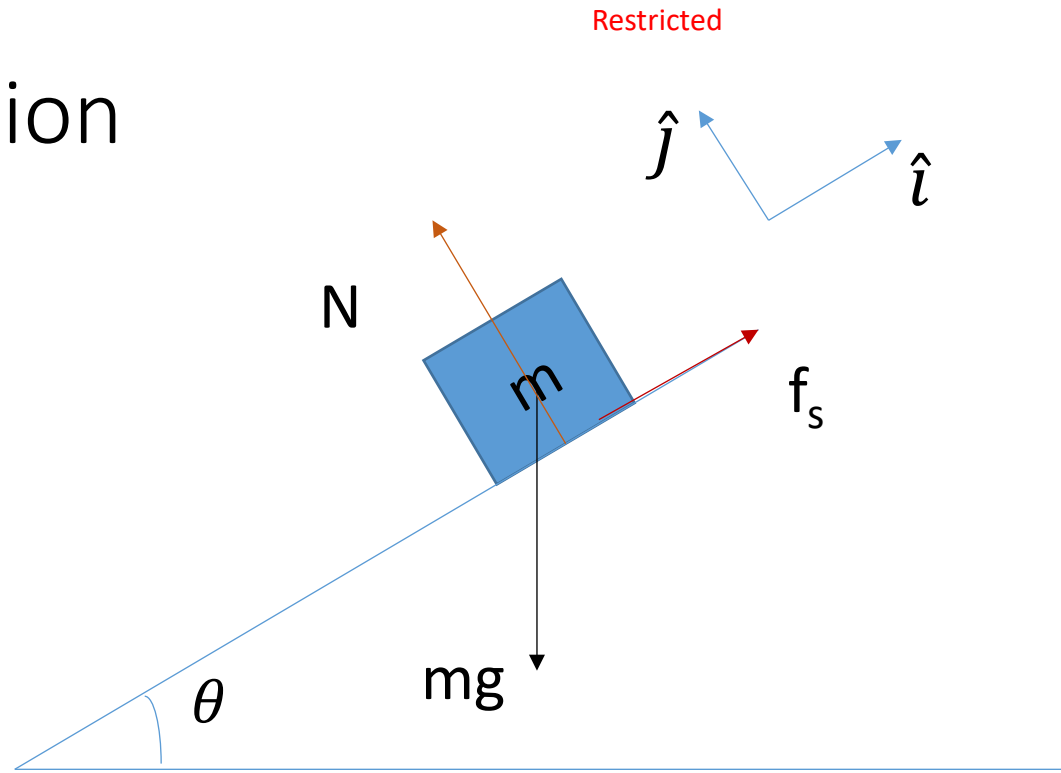
https://web.njit.edu/~tyson/P111_chapter4.pdf

Example



Draw free-body diagram for m which is placed on a slope, and it is at rest. Assume the surface is not frictionless. Find the net force acting on m .

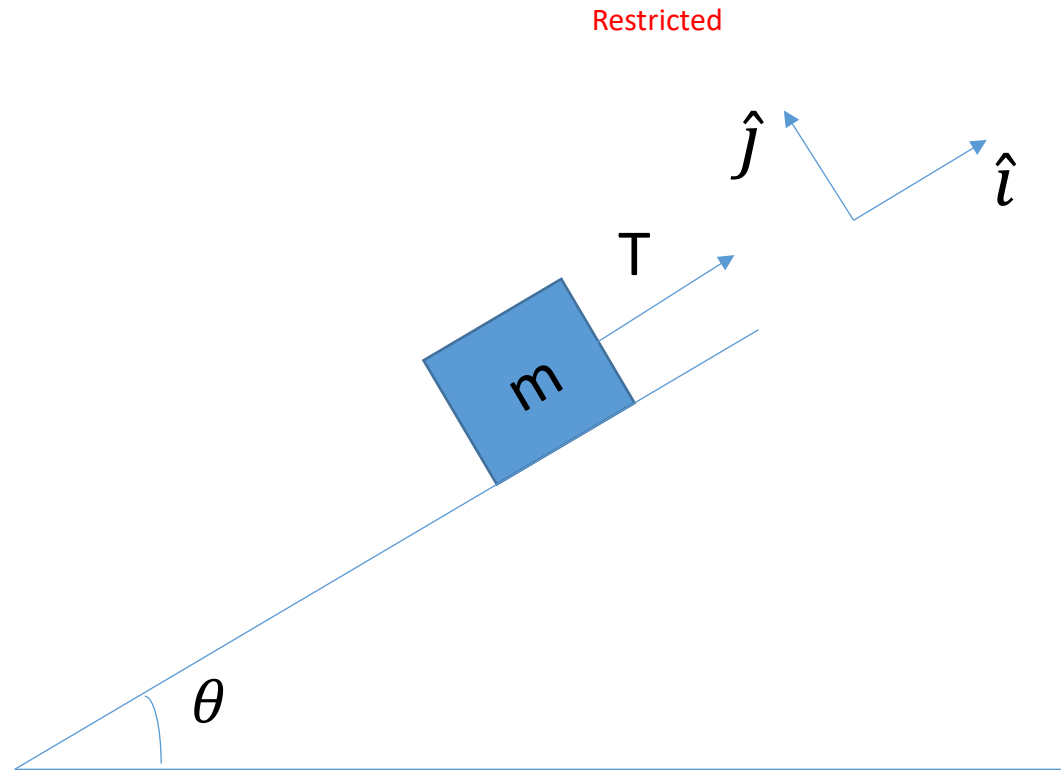
Example: Solution



Draw free-body diagram for m which is placed on a slope, and it is at rest. Assume the surface is not frictionless. Find the net force acting on m .

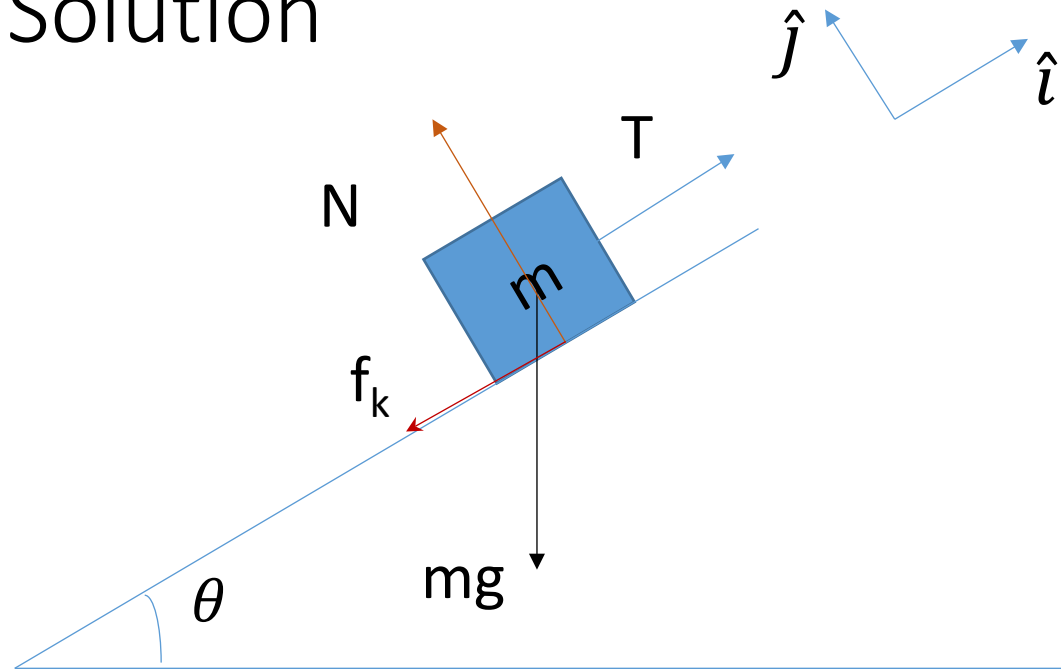
$$\begin{aligned}\vec{F}_N &= mg \sin \theta (-\hat{i}) + mg \cos \theta (-\hat{j}) + f_s \hat{i} + N \hat{j} \\ &= (f_s - mg \sin \theta) \hat{i} + (N - mg \cos \theta) \hat{j}\end{aligned}$$

Case Problem



Draw free-body diagram for m which is placed on a slope, and it is moving upwards. Assume the surface is not frictionless. Find the net force acting on m .

Case Problem: Solution

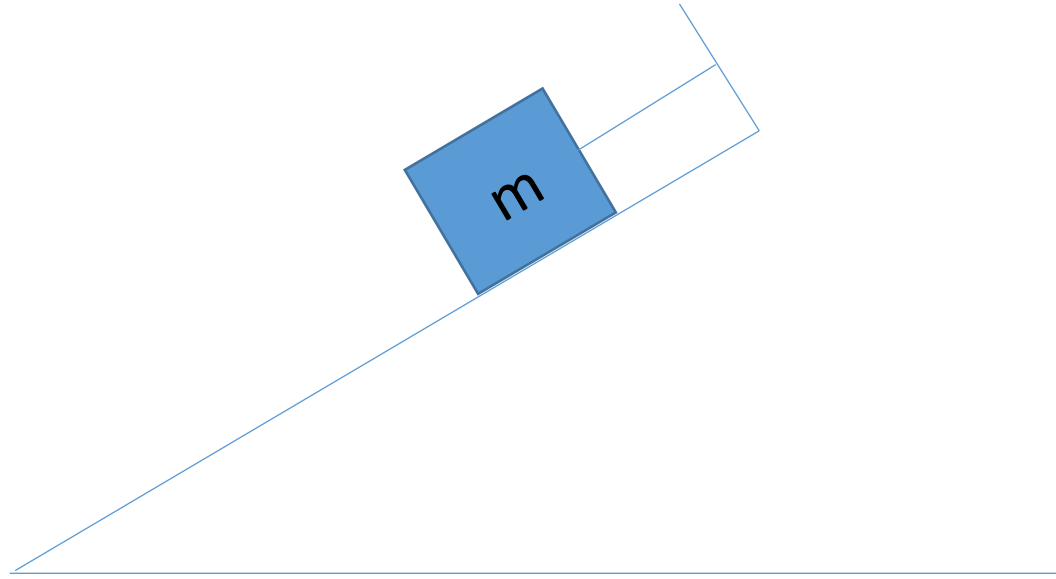


Draw free-body diagram for m which is placed on a slope, and it is moving upwards. Assume the surface is not frictionless. Find the net force acting on m .

$$\begin{aligned}\vec{F}_N &= mg \sin \theta (-\hat{i}) + mg \cos \theta (-\hat{j}) + f_k (-\hat{i}) + N\hat{j} + T\hat{i} \\ &= (T - mg \sin \theta - f_k) \hat{i} + (N - mg \cos \theta) \hat{j}\end{aligned}$$

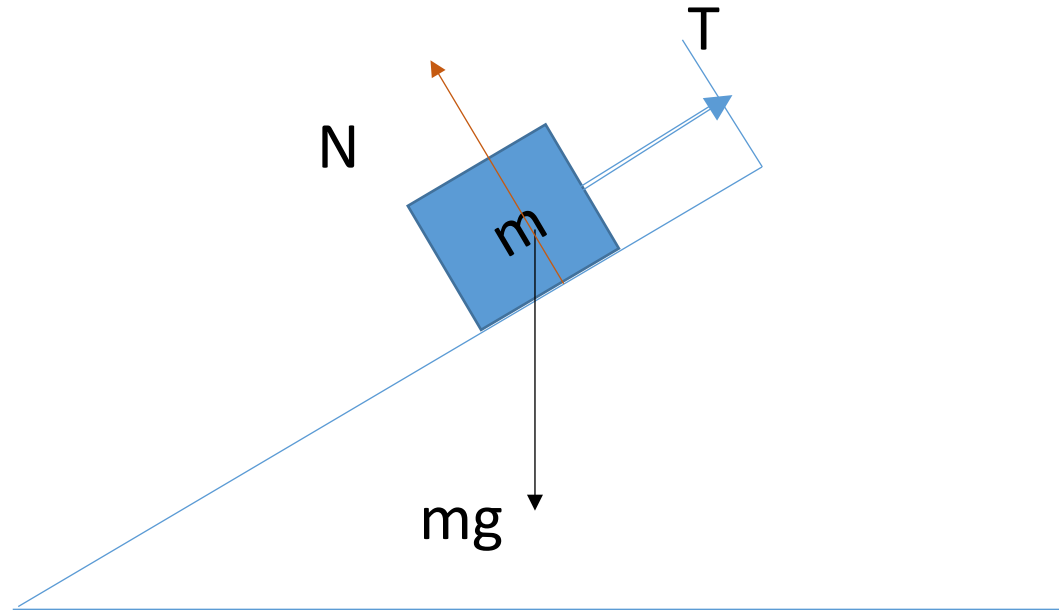
Off-class practice question(s)

Case Problem



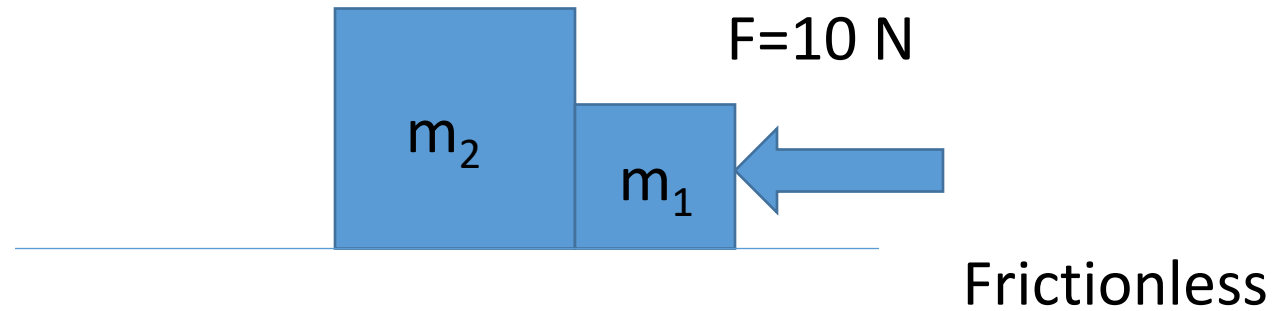
Draw free-body diagram for m which is placed on a slope and connected to a wall via a string, and it is at rest. Assume the surface is frictionless.

Case Problem: Solution



Draw free-body diagram for m which is placed on a slope and connected to a wall via a string, and it is at rest. Assume the surface is frictionless.

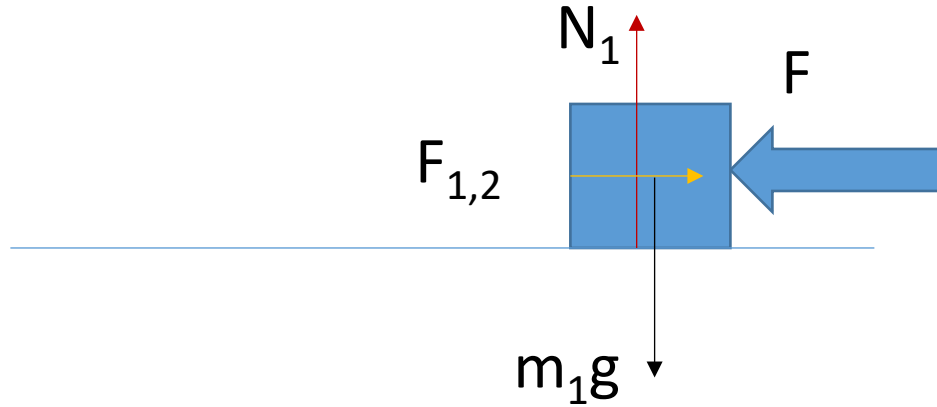
Case Problem



Draw free-body diagram for m_1 and m_2 .

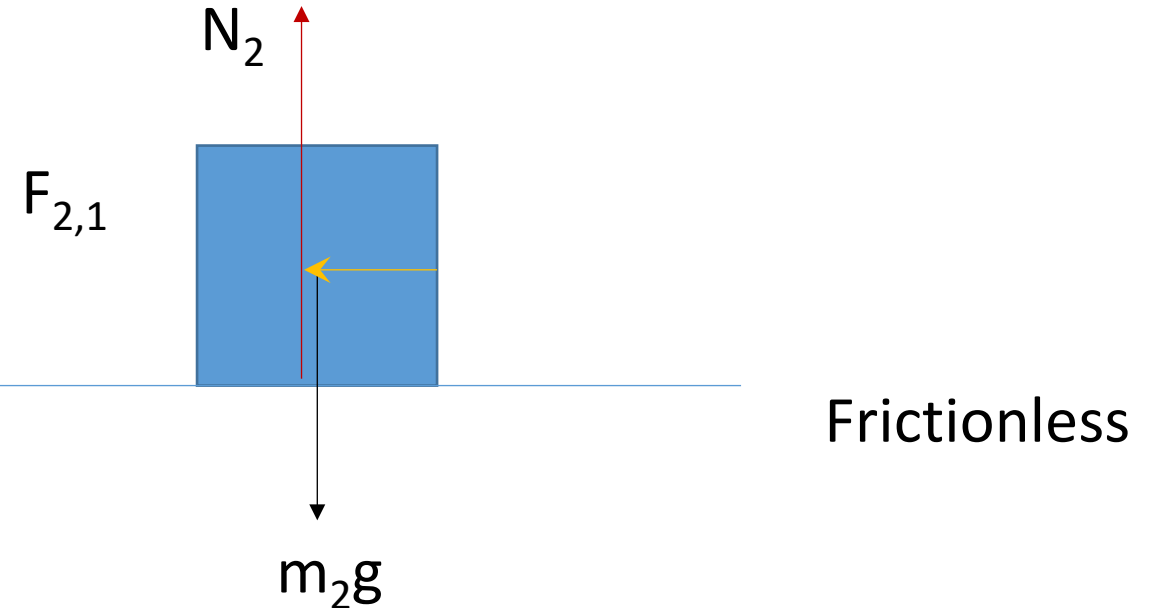
- Find the net force acting on m_1 , m_2 , separately.
- Find the net force acting on the system of m_1 and m_2 .

Case Problem: Solution



$$\begin{aligned}\vec{F}_{Net1} &= F(-\hat{i}) + F_{1,2}\hat{i} + N_1\hat{j} + m_1g(-\hat{j}) \\ &= (F_{1,2} - F)\hat{i} + (N_1 - m_1g)\hat{j}\end{aligned}$$

$$\begin{aligned}\vec{F}_{Net2} &= F_{2,1}(-\hat{i}) + N_2\hat{j} + m_2g(-\hat{j}) \\ &= F_{2,1}(-\hat{i}) + (N_2 - m_2g)\hat{j}\end{aligned}$$



Case Problem: Solution

$$\vec{F}_{Net1} = F(-\hat{i}) + F_{1,2}\hat{i} + N_1\hat{j} + m_1g(-\hat{j}) = (F_{1,2} - F)\hat{i} + (N_1 - m_1g)\hat{j}$$

$$\vec{F}_{Net2} = F_{2,1}(-\hat{i}) + N_2\hat{j} + m_2g(-\hat{j}) = F_{2,1}(-\hat{i}) + (N_2 - m_2g)\hat{j}$$

$$\vec{F}_{Net} = \vec{F}_{Net1} + \vec{F}_{Net2}$$

$$= (F_{1,2} - F_{2,1} - F)\hat{i} + (N_1 - m_1g + N_2 - m_2g)\hat{j}$$